

THE GEOMETRY OF SPACE

Cosmologists base their ideas on the fate of the universe partly on mathematical models. These indicate that, depending on its mass-energy density, the universe has three possible geometries, each with a different space-time curvature that can be represented by a 2-D shape. Before the discovery of dark energy, there was a correspondence between these geometries and the fate of the universe. A positively curved or "closed" universe was envisaged to end in a Big Crunch and a negatively curved or "open" universe in a Big Chill. A "flat" universe would also end in a Big Chill, but one in which the universe's expansion eventually slows to a virtual standstill. With the discovery of dark energy, the correspondence no longer holds. If dark energy remains constant in intensity, any type of universe may expand forever. If dark energy is capable of reversing, any type of universe could end in a Big Crunch. Currently, the most favored view is that the universe is flat and will undergo an accelerating expansion.

A cataclysmic "Big Rip" scenario, in which increasing dark energy tears the universe apart, is less likely.

Big Crunch

FLAT UNIVERSE

If the density of the universe is exactly on a critical value, it is "flat." In a flat universe, parallel lines never meet. The 2-D analogy is a plane. The universe is thought to be flat or nearly flat.

CLOSED UNIVERSE

If the universe is denser than a critical value, it is positively curved or "closed" and is finite in mass and extent. In such a universe, parallel lines converge. The 2-D analogy is a spherical surface.

OPEN UNIVERSE

If the universe is less dense than a critical value, it is negatively curved or "open" and infinite. The 2-D analogy of such a universe is a saddle-shaped surface on which parallel lines diverge.

A COLD DEATH

If the universe peters out in a Big Chill, its death will take a long time. Over the next 10^{12} (1 trillion) years, galaxies will exhaust their gas in forming new stars. About 10^{25} (10 trillion trillion) years in the future, most of the universe's matter will be locked up in star corpses such as black holes and burned-out white dwarfs circling and falling into the supermassive black holes at the centers of galaxies. At 10^{32} (1 followed by 32 zeros) years from now, protons will start decaying to radiation (photons), electrons, positrons, and neutrinos. All matter not in black holes will fall apart. After another 10^{67} years, black holes will start evaporating by emitting particles and radiation, and in about 10^{100} years,

even supermassive black holes will evaporate. The utterly cold, dark universe then will be nothing but a diffuse sea of photons and elemental particles.

FATE OF GALAXIES

A trillion years from now, the universe will contain just old, fading galaxies. All their gas and dust will be used up, and most of the stars will be dying.

BIG CRUNCH

In this version of doomsday, all matter and energy collapse to an infinitely hot, dense singularity in a reverse of the Big Bang. This scenario currently looks the least likely unless the effect of dark energy reverses in future. Even if it did happen, the earliest it could do so would be tens of billions of years from now.

present day

BIG BANG

FINAL SURVIVORS

At a very distant stage of the Big Chill, all the universe's matter, even that in black holes, will have decayed or evaporated to radiation. Apart from some very long-wavelength photons, the only constituents of the universe will be neutrinos, electrons, and positrons.

photon

neutrino